

Analysis of Amino Acid Sequences of Penicillin-Binding Proteins 1a, 2b, and 2x in Invasive *Streptococcus pneumoniae* Nonsusceptible to Penicillin Isolated from Children in India

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Penicillin-binding proteins are the primary targets for beta lactam drugs, which are main stay of treatment for *Streptococcus pneumoniae*. The emergence of increased penicillin resistance in meningeal isolates of *S. pneumoniae* in India is alarming. With this background, we aimed to analyze the *pbp* gene mutations of penicillin nonsusceptible pneumococcal (PNSP) isolates from within India and their association with international clones. A total of 32 PNSP invasive isolates with a penicillin minimal inhibitory concentrations (MIC) of ≥ 0.12 $\mu\text{g/mL}$ were subjected to PCR and sequencing for multilocus sequence typing and the *pbp* genes (*pbp2b*, *pbp2x*, and *pbp1a*). The *S. pneumoniae* R6 susceptible strain was used as the reference for the comparison analyses. In the majority of the present study isolates, amino acid substitutions were only seen in one of the three active sites of one of the three *pbp* genes. Thus, *pbp* genes in the absence of the major substitutions usually associated with penicillin resistance combined with mosaicism in *pbp1a* resulted in a slight increase in the penicillin MIC to between 0.06 and 2.0 $\mu\text{g/mL}$, which according to meningeal break point denote resistance. Clonal analyses revealed that the emergence of PNSP in India is due to the gradual expansion of the resistant clones CC320, CC230, and CC63.

Keywords: Penicillin nonsusceptible *S. pneumoniae*, penicillin-binding proteins, invasive, India

Introduction

THE WORLDWIDE EMERGENCE of pneumococcal resistance to penicillin poses the greatest threat to the management of pneumococcal infections.¹ Although penicillin nonsusceptible pneumococci (PNSP) first emerged in the 1960s, the first report from India was in 1999 with the prevalence only being 1.3%.² However, penicillin resistance among pneumococcal meningeal isolates in India increased from 9.5% in 2008 to 42.8% in 2016, while cefotaxime nonsusceptibility increased from 4.7% in 2008 to 28.5% in 2016.³ Penicillin-binding proteins (PBP) are the primary target for beta lactam drugs, including penicillin which thereby inhibits the final steps of cell wall synthesis.⁴

Pneumococcal resistance to penicillin is caused by altered *pbp* genes, *pbp 2x*, *2b*, and *1a*, mainly in the active sites which thereby reduce the affinity of beta lactam antibiotics.^{5–7} Reduced affinity is due to mutations that generate mosaic *pbp* genes through recombinational events with closely related commensal Streptococcal spp. (*Strepto-*

coccus mitis group) or other penicillin-resistant pneumococci.⁸ Several epidemiologic studies on penicillin-resistant pneumococci have shown clones with these resistant genes disseminate under selective pressures rather than the emergence of a new resistant strain.^{9–12} There is a lack of published literature on the molecular epidemiology of PNSP isolates and their related PBP mutation profiles in India. The aim of the present study is to screen for amino acid (aa) variations in the transpeptidase domain (TPD) of the *pbp2x*, *pbp2b*, and *pbp1a* genes of invasive PNSP isolates from India and to characterize them using multilocus sequence typing (MLST).

Materials and Methods

Characterization of bacterial strains and antimicrobial susceptibility testing

Invasive *Streptococcus pneumoniae* (blood, CSF, and sterile body fluids) were isolated between January 2012 and December 2014, from children <5 years of age admitted to

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Christian Medical College in Vellore (India). A total of 32 invasive isolates (meningeal 18, nonmeningeal 14) of PNSP (penicillin minimal inhibitory concentrations [MIC] ≥ 0.12 $\mu\text{g/mL}$) were included in the study. *S. pneumoniae* was identified using standard methods such as optochin susceptibility and bile solubility testing.¹³ The MIC of penicillin, erythromycin, cefotaxime, linezolid, vancomycin, levofloxacin, and cotrimoxazole were determined using *E* test and interpreted as per CLSI guidelines.¹⁴ Serotyping of the isolates was performed by both the Quellung method using pneumococcal antiserum (Statens Serum Institut, Denmark) as previously described¹³ and customized sequential multiplex PCR.¹⁵

Multilocus sequence typing

MLST was performed for the 32 PNSP isolates and analyzed as described previously by Enright and Spratt.¹⁶ The sequence types (STs) were derived from the PubMLST database, and the new allele profile and new allele numbers were obtained by submitting the sequences to the database curators. The genetic relatedness of these isolates to the international Pneumococcal Molecular Epidemiology Network (PMEN) clones was deciphered from the PubMLST database. Their relationship to the global STs was obtained using goeBURST analysis software against the whole ST profile in the PubMLST database as on February 16, 2018.

PBP sequencing

TPD of the *pbp2x*, *pbp2b*, and *pbp1a* gene fragments was amplified and sequenced using primers described earlier.^{17,18} The amplified PCR products were sequenced using the ABI 3500 Genetic Analyzer (Applied Biosystems, Thermo Fisher Scientific Company, Waltham, MA).

Bioinformatic tools and analysis

The complete nucleotide sequences of the study isolates were translated to the amino acid sequences for three genes namely *pbp2x*, *pbp2b*, and *pbp1a* and compared with the reference strain *S. pneumoniae* R6 (sequence available at GenBank accession numbers: *pbp1a* M90527; *pbp2b* X16022; *pbp2x* X16367) using Clustal alignment software.

Results

The penicillin MIC values of the 32 pneumococcal isolates were ranged between 0.12 and 4 $\mu\text{g/mL}$. Among the total of 32 PNSP isolates, majority ($n=21$) had MIC ranged between 0.12 and 1.0 $\mu\text{g/mL}$, out of which 10 are meningeal isolates with low-level MIC between 0.12 and 0.25 $\mu\text{g/mL}$. The major serotypes observed were 14, 6B, 6A, 19F, 19A, and 23F. Most of the PNSP isolates showed additional nonsusceptibility to erythromycin ($n=21$) and cefotaxime ($n=9$). All were susceptible to linezolid, vancomycin, and levofloxacin and resistant to cotrimoxazole. The serotypes, MICs, and number of amino acid substitutions observed in the conserved motifs of PBP1a, PBP2b, and PBP2x are detailed in Table 1. The amino acid changes present in the transpeptidase region of the three PBPs compared to the reference are detailed in the Supplementary Tables S1–S3. Majority of the isolates belonged to clonal complexes CC320/GroupCC0 ($n=8$), followed by CC63/CCgroup1

($n=7$) and CC230/CCgroup6 ($n=5$). The other CCs are CC473/CCgroup5 ($n=2$) and one isolate each of ST9428, 13055, 10812, 13077, 874, 13160, 9430, 9432, 9422, and 9424. The major STs were 63 and 3135.

Variation in the TPD of the *pbp* genes

Mutations at positions in or close to each of the three (SXXN, SXN, and KXG) conserved motifs for the three *pbp* genes were identified and analyzed. The three conserved motifs are the following: for PBP2b—S386VVK, S443SNT, and K615TG; PBP2x—S337TMK, S395SN, and K547SG; and PBP1a—S370TMK, S428RN, and K557TG.

PBP1a

Active site changes were only seen in S370TMK motif (T371A-56%). P432T substitution next to the active site of S428RN motif were seen in 63% ($n=20$) isolates and no changes were seen in K557TG motif. In PBP1a, adjacent substitutions in four positions N₅₇₄TGY replacing T₅₇₄ SQF were seen in 23 isolates (72%). E388D substitution was present in all the isolates (see Table 1 and Supplementary Table S1).

PBP2b

Majority of the amino acid alterations in PBP2b of penicillin intermediate and resistant isolates occurred within a ± 250 bp region between Thr408 and Thr488. These areas are located between the two active site motifs S386VVK and S443SNT. T446A, major substitution next to the active site S443SN, was seen in 84% ($n=27$) isolates. Four prominent amino acid substitutions such as T446A, E476G, T489A, and V538D were present in $\sim 73\%$ of the isolates. Of these, V538D is a new substitution seen in 75% of the isolates and has not reported earlier. No changes were observed in the other active sites KTG and SVVK (Table 1 and Supplementary Table S2).

PBP2x

Two commonly altered sites in PBP2x that confer resistance to penicillin and/or cefotaxime are T338(A/G/P/S) in the active site STMK and Q552E next to site KSG. T338A and Q552 E substitutions were found only in 38% and 16% of this study isolates.

Instead, amino acid change L565V/S was seen in 72% among study isolates. Another mutation H394Y next to S395SN motif was also observed in six isolates. L565S when present as a single mutation in three of the isolates showed increase MIC of 0.25 and 0.5 $\mu\text{g/mL}$. A369V substitution, which is a resistance determinant, was present in 28% of the isolates (Table 1 and Supplementary Table S3).

Phylogenetic analysis of PBP2x sequences

Phylogenetic analyses of *pbp2x* sequences from the study isolates and the reference sequences of *Streptococcus oralis* M3 (X78217), *Streptococcus infantis* JR (KY292549), and *S. mitis* B6 (AJ002288) gave rise to four clusters (Fig. 1). The *pbp2x* gene was chosen for the analyses because it showed a high degree of mosaicism with the study isolates

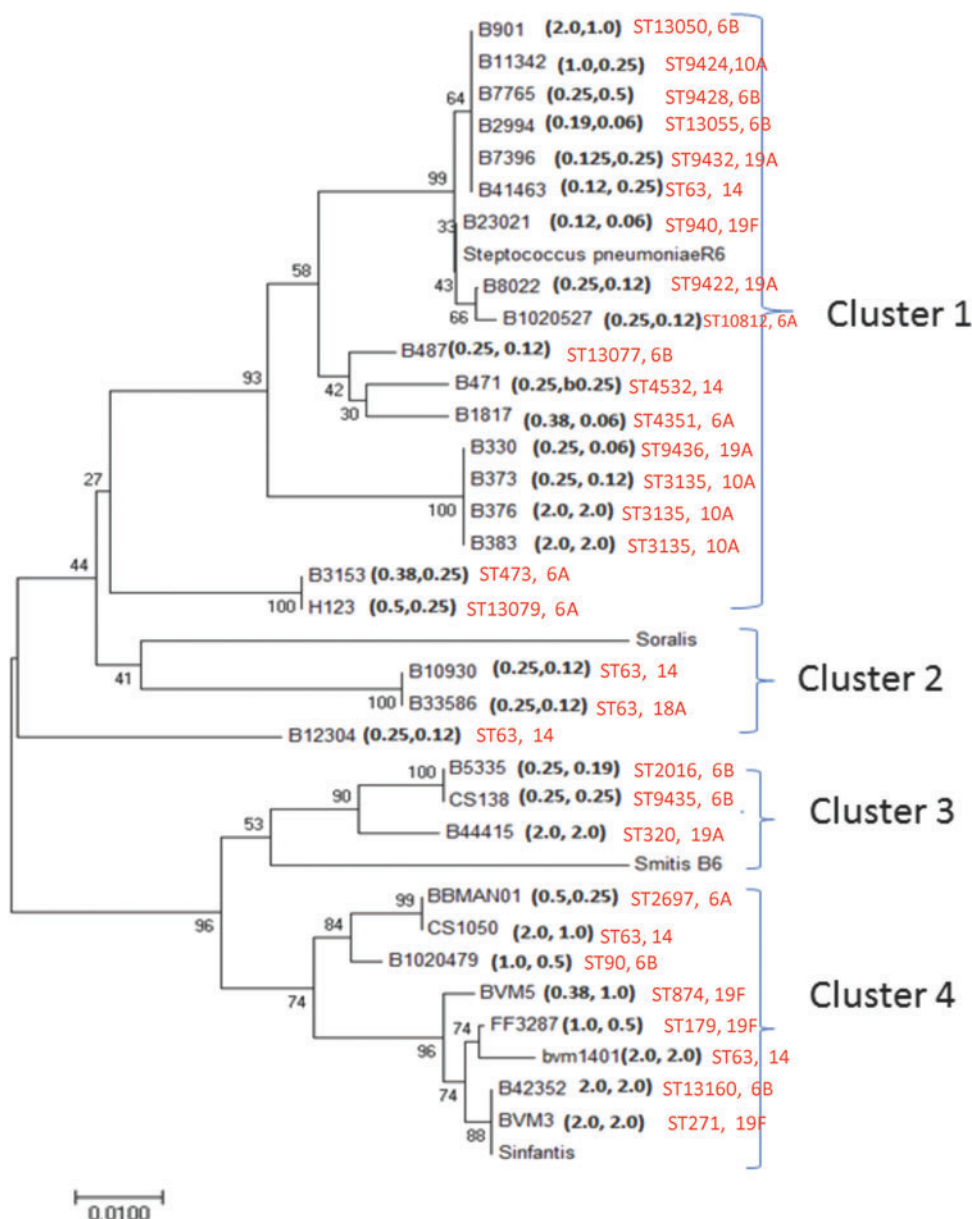
TABLE 1. SEQUENCE TYPES, SEROTYPES MINIMAL INHIBITORY CONCENTRATIONS, AND THE TOTAL NUMBER OF AMINO ACID CHANGES ALONG WITH THE MUTATIONS IN THE CONSERVED MOTIFS OF THE PENICILLIN NONSUSCEPTIBLE PNEUMOCOCCAL ISOLATES

Isolate ID	Serotype	ST/C Group	PEN MIC	CTX MIC	N (%) of altered amino acids				Amino acid mutations in the conserved motifs						
					PBP2b		PBP2x		PBP1a		PBP2b		PBP2x		PBP1a
					345-646 ^a	271-616 ^a	328-584 ^a	SVVK	SSNT	KTGTA	STMK	AHSSNV	LKSGT	STMK	SRNVP
CS138(M)	6B	9435/0	0.25	0.19	8 (2.6)	38 (10.9)	22 (8.5)	---	---	---	---	---	---	---	---
B5335(M)	6B	2016/0	0.25	0.25	7 (2.3)	39 (11.2)	22 (8.5)	---	---	---	---	---	---	---	---
BMA1(M)	6A	2697/0	0.5	0.25	32 (10.6)	36 (11.9)	23 (8.9)	---	---	---	---	---	---	---	---
B1020479(NM)	6B	90/0	1	0.5	15 (5.2)	35 (11.5)	25 (9.7)	---	---	---	---	---	V---	---	---
FF3287(NM)	19F	179/0	1	0.5	16 (5.3)	42 (13.9)	24 (9.3)	---	---	---	A-	---	---	---	---
B901(NM)	6B	13057/0	2	1	22 (7.3)	1 (0.3)	25 (9.7)	---	---	---	---	---	---	---	---
BVM 3(M)	19F	271/0	2	2	12 (4)	49 (16.2)	24 (9.3)	---	---	---	A-	---	---	---	---
B44415(NM)	19A	320/0	2	2	37 (12.2)	47 (15.5)	24 (9.3)	---	---	---	---	L---	---	---	---
B330 (NM)	19A	9436/6	0.25	0.06	4 (1.32)	18 (5.9)	26 (10.1)	---	---	---	---	L---	---	---	---
B 373(M)	10A	3135/6	0.25	0.12	33 (10.9)	18 (5.9)	17 (6.6)	---	---	---	---	---	---	---	---
B1817(NM)	6A	4351/6	0.38	0.38	12 (3.4)	12 (3.5)	29 (11.2)	---	---	---	---	---	V---	A-	---
B 376(M)	10A	3135/6	2	2	33 (10.9)	17 (4.9)	24 (9.3)	---	---	G	A-	---	V---	S-	---
B383(M)	10A	3135/6	2	2	34 (11.2)	18 (5.2)	24 (9.3)	---	---	---	A-	---	V---	S-	---
B41463(M)	14	63/1	0.125	0.25	9 (3)	19 (5.4)	20 (7.7)	---	---	---	---	---	---	---	---
B12304(NM)	14	63/1	0.25	0.12	10 (3.3)	31 (8.9)	24 (7.9)	---	---	---	---	---	---	---	---
B10930(NM)	14	63/1	0.25	0.12	11 (3.6)	29 (8.4)	25 (9.3)	---	---	---	A-	---	V---	S-	---
B 471(M)	14	4532/1	0.25	0.25	9 (3)	10 (2.9)	25 (9.3)	---	---	---	A-	---	V---	S-	---
B33586(NM)	18A	63/1	0.25	0.12	10 (3.3)	29 (8.4)	17 (6.6)	---	---	---	A-	---	V---	A-	---
CS1050(M)	14	63/1	2	1	11 (3.6)	36 (10.4)	24 (9.3)	---	---	---	---	---	---	---	---
B 1401(M)	14	63/1	2	2	32 (10.6)	43 (12.4)	29 (11.2)	---	---	---	---	---	---	---	---
B3153(NM)	6A	473/5	0.25	0.25	12 (4)	25 (7.22)	10 (3.8)	---	---	---	---	---	---	A-	---
H123(NM)	6A	13079/5	0.5	0.25	9 (3)	25 (7.22)	9 (3.5)	---	---	---	---	---	---	S-	---
B23021(NM)	19F	9430/S	0.12	0.06	12 (3.9)	0	2 (0.8)	---	---	---	---	---	---	---	---
B7396(NM)	19A	9432/S	0.12	0.25	12 (3.9)	1 (0.2)	3 (1.2)	---	---	---	---	---	---	---	---
B2994(M)	6B	13055/S	0.25	0.06	6 (2)	29 (8.4)	17 (6.6)	---	---	A-	---	---	---	---	---
B7765(M)	6B	9428/11	0.25	0.5	1 (0.3)	1 (0.2)	1 (0.3)	---	---	---	---	---	---	---	---
B8022(NM)	19A	9422/S	0.25	0.12	3 (1)	1 (0.2)	24 (8.9)	---	---	---	---	---	---	S-	---
B1020527 (M)	6A	10812/424	0.25	0.12	2 (0.7)	2 (0.6)	4 (1.2)	---	---	---	---	L---	---	---	---
B 487(M)	6B	13077/S	0.25	0.12	13 (4.3)	10 (2.9)	25 (9.3)	---	---	---	---	---	---	---	---
BVM5(M)	19F	874/27	0.5	1.01	12 (4)	42 (12.1)	28 (10.8)	---	---	---	---	---	V---	A-	---
BB11342(NM)	10A	9424/S	1.0	0.25	1 (0.3)	1 (0.2)	1 (0.3)	---	---	---	---	---	---	---	---
B42352 (M)	6B	13160/S	4	2	12 (4)	45 (13)	24 (8.5)	---	---	A-	---	---	V---	S-	---

^aTPD region.

CTX, Cefotaxime; M, Meningeal; MIC, minimal inhibitory concentrations; NM, Non Meningeal; PEN, Penicillin; S, Singleton; ST, sequence types.

FIG. 1. Phylogenetic analyses of *pbp2x* genes. Neighbor-joining tree was generated from 32 PBP sequences of study isolates and 4 reference strains using MEGA (V 7.0) software. Bootstrap values of 1,000 replications are shown. Cluster 1–4 were defined according to the reference *Streptococcus pneumoniae* R6 (Cluster 1), *Streptococcus oralis* M3 (cluster 2), *Streptococcus mitis* B6 (cluster 3), and *Streptococcus infantis* JR (Cluster 4). The bar refers to genetic divergence as calculated by the MEGA software. Penicillin and cefotaxime MIC values are within brackets next to the branch labels. The sequence type and serotype are denoted in red color text at the end of branch labels. MIC, minimal inhibitory concentrations; PBP, penicillin-binding proteins. Color images are available online.



compared to the other two genes. The *pbp2x* gene of the study isolates with an increased MIC belonged to one of the reference clusters of either *S. infantis* JR, *S. mitis* B6, or *S. oralis* M3.

PBP sequences in relationship to clonal complex

The 32 PNSP study isolates showed 25 STs (12 new STs) out of 10 serotypes, which were distributed in 7 clonal complexes and 7 singletons by goeBURST analysis against whole STs in the PubMLST database. The PMEN clones associated with the study PNSP isolates are Taiwan^{19F}-14 (ST236), Denmark¹⁴-32 (ST230), Sweden^{15A}-25 (ST63), Spain^{6B}-2 (ST90), and Portugal^{19F}-21 (ST177). (Fig. 2). Table 2 shows the related clonal complexes within the PNSP isolates. The common amino acid substitutions observed between CCs are explained in Supplementary Table S4.

Discussion

Variations in PBP genes, *pbp2b*, *2x*, and *1a*, are the major mechanism of penicillin resistance in *S. pneumoniae*. The variations are due to recombination of these genes between commensal streptococcal species. Previous studies reported that *pbp2b* gene mutations are selected by penicillin, *pbp2x* gene mutations are selected by cefotaxime and oxacillin, whereas changes in the *pbp1a* gene are responsible for high-level resistance in both penicillin and cefotaxime.¹⁹ In addition, amino acid mutations in *pbp2b* only increase the MIC values up to twofold for penicillin compared to amino acid mutations in *pbp2x* increase the MIC value up to 1.5- and 3-fold for cefotaxime. A mosaic *pbp1a* gene has a greater effect on penicillin resistance compared to cefotaxime resistance.¹⁹

In this study, the amino acid substitutions caused by mutation in the three *pbp* genes are similar to those

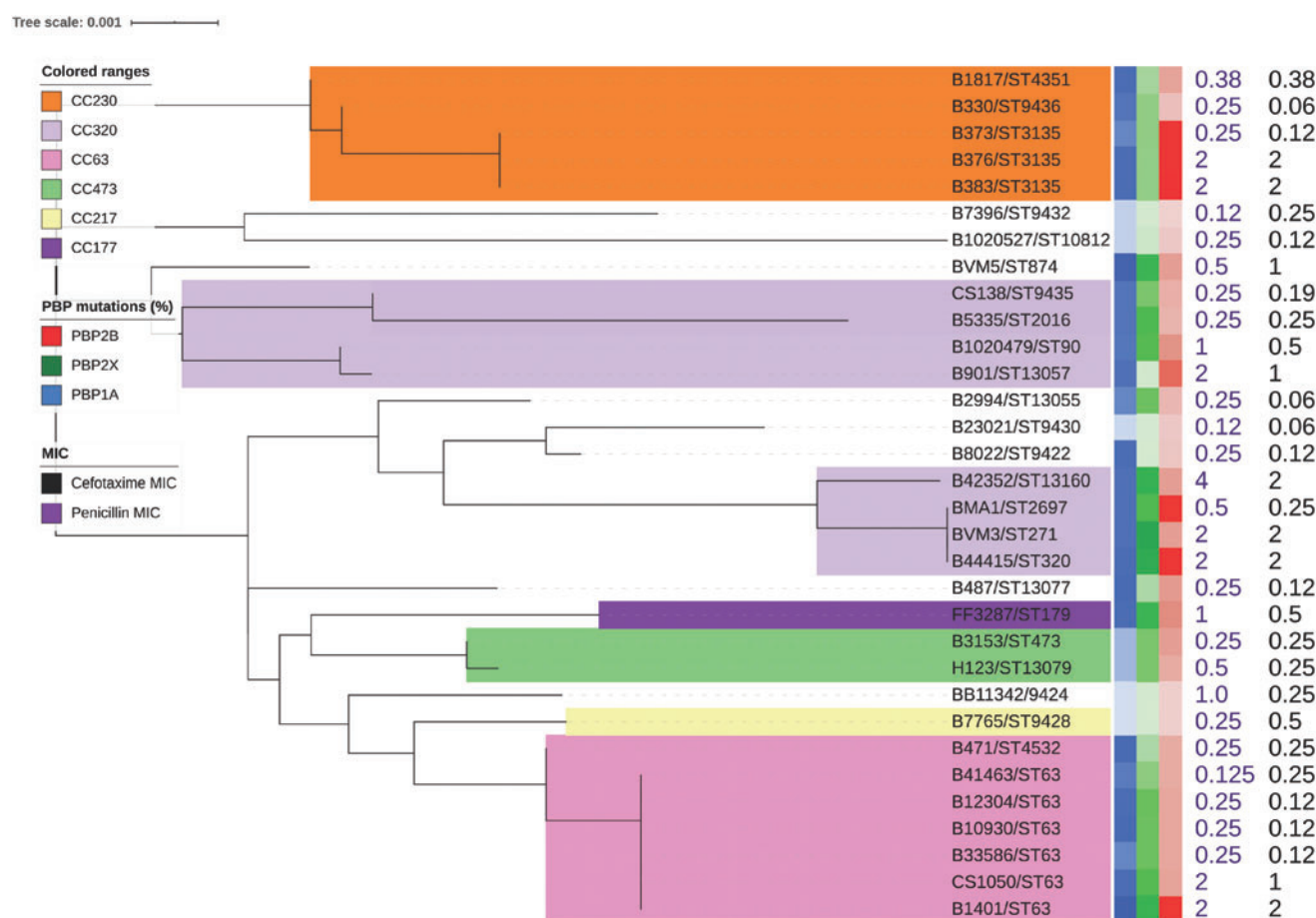


FIG. 2. Maximum likelihood tree created using MEGA v 7.0 using concatenated sequences of MLST genes of the sequence types of the penicillin nonsusceptible isolates. The tree is labeled using iTOL software. The clades related to PMEN CC are shaded. The heatmap strip indicates the percentage of amino acid mutations in the respective PBP. The MIC values of penicillin and cefotaxime are mentioned as text. PMEN, Pneumococcal Molecular Epidemiology Network; MLST, multilocus sequence typing. Color images are available online.

previously reported ones. The difference was in the absence of major mutation combinations that is required for high-level resistance.^{19–21}

The maximum MIC level of PNSP reported in IPD cases in India is relatively low with MIC of 2.0 µg/mL compared to high MIC of >8.0 µg/mL reported in many countries.^{3,22,23} Previous studies on PBPs suggest increase in the level of resistance along with the degree of amino acid variations. Our analysis on PBP 2x, 2b, and 1a amino acid sequences of invasive pneumococcal isolates revealed that among the three PBPs, the overall percentage of amino acid variation in the TPD is very less and similar in PBP2x and PBP2b (21%) compared to PBP1a (14%). However, isolates with high number of aa variations in TPD domain were more in PBP2x compared to PBP2b. This may be because of PBP2x, the first target to undergo mutation under antibiotic pressure.^{24–26} Those isolates with less aa variations in PBP2b had numerous aa variations in corresponding PBP2x and PBP1a TPD domain sequences.

None of the isolates had aa changes in the three active sites of PBP2b. Only active site STMK in PBP2x [T338(A/G/P/S) and PBP1a (T371A/S)] had mutations which is seen in isolates with penicillin MIC more than or

equal to 0.25 µg/mL and cefotaxime MIC ≥0.06 µg/mL, but these mutations were not present in all the isolates. This is similar to findings reported by Zhou *et al.* and Diawara *et al.*^{22,27} Another substitution H394L, before the second conserved motif of PBP2x, was found in isolates with penicillin MIC >0.25 µg/mL, but not associated with T338A mutation as reported by Granger *et al.*²⁸ Observation of N417 K mutation even in isolates with MIC 0.25 µg/mL, differed from findings of Granger *et al.*²⁸ They reported these mutations only in isolates with MIC of 0.5 or 1.0 µg/mL. L546V substitution associated with high-level beta lactam resistance is also found in isolates with MIC of 0.25 µg/mL similar to findings of Granger *et al.*²⁸ The important mutation T550A in *pbp2x* gene, which confers high-level cefotaxime resistance in laboratory mutants, is absent in study isolates. This is in concordance with the reported difference in resistance observed between clinical strains and laboratory mutants. In study from Turkey by Biçmen *et al.*,²⁹ substitutions T338A, Q552E, and S571P together reduce the binding affinity of penicillin and cefotaxime by 80%. However, these substitutions were not observed together in any of the study isolates. Similarly, M339F substitution, which is associated with penicillin MIC >4 µg/mL

TABLE 2. DISTRIBUTION OF SEQUENCE TYPES TO CC GROUPS AND THEIR RELATIONSHIP TO THE PNEUMOCOCCAL MOLECULAR EPIDEMIOLOGY NETWORK CLONES AND THE CORRESPONDING NUMBER OF AMINO ACID CHANGES

CCgroup/FG	ST	Relation to PMEN clone	Year of isolation	Serotype	Pen MIC	aroE	gdh	Gki	recP	spi	xpt	ddl	N (%) of altered amino acids		
													PBP2b 345–646 ^a	PBP2x 271–616 ^a	PBP1a 328–584 ^a
CC0/320	9435	SLV of ST2016	2012	6B	0.25	7	410	1	2	6	1	19	8 (2.6)	38 (10.9)	22 (8.5)
CC0/320	2016	SLV of ST 9435	2013	6B	0.25	7	57	1	2	6	1	19	7 (2.3)	39 (11.2)	22 (8.5)
CC0/320	2697	SLV of ST 271,320	2013	6A	0.5	4	16	19	15	6	20	252	32 (10.6)	36 (11.9)	23 (8.9)
CC0/320	90	Spain ^{6B-2} ST 90	2014	6B	1	5	6	1	2	6	3	4	15 (5.2)	35 (11.5)	25 (9.7)
CC0/320	179	SLV of ST177 Portugal ^{19F-21}	2014	19F	1	7	14	40	12	1	1	14	16 (5.3)	42 (13.9)	24 (9.3)
CC0/320	13057	DLV of ST 90 Spain ^{6B-2}	2012	6B	2	5	6	207	2	6	3	9	22 (7.3)	1 (0.3)	25 (9.7)
CC0/320	271	SLV of ST236 Taiwan ^{19F-14}	2012	19F	2	4	16	19	15	6	20	26	12 (4)	49 (16.2)	24 (9.3)
CC0/320	320	DLV of ST 236 Taiwan ^{19F-14}	2014	19A	2	4	16	19	15	6	20	1	37 (12.2)	47 (15.5)	24 (9.3)
CC6/230	9436	DLV of ST230 Denmark ¹⁴⁻³²	2013	19A	0.25	12	10	2	17	6	22	5	4 (1.32)	18 (5.9)	26 (10.1)
CC6/230	3135	DLV of ST230 Denmark ¹⁴⁻³²	2014	10A	0.25	5	19	2	17	6	22	9	33 (10.9)	18 (5.9)	17 (6.6)
CC6/230	4351	SLV of ST230 Denmark ¹⁴⁻³²	2013	6A	0.38	12	5	2	17	6	22	14	12 (3.4)	12 (3.5)	29 (11.2)
CC6/230	3135	DLV of ST230 Denmark ¹⁴⁻³²	2014	10A	2	5	19	2	17	6	22	9	33 (10.9)	17 (4.9)	24 (9.3)
CC6/230	3135	DLV of ST230 Denmark ¹⁴⁻³²	2014	10A	2	5	19	2	17	6	22	9	34 (11.2)	18 (5.2)	24 (9.3)
CC1/63	63	Sweden ^{15A-25}	2013	14	0.125	2	5	36	12	17	21	14	9 (3)	19 (5.4)	20 (7.7)
CC1/63	63	Sweden ^{15A-25}	2014	14	0.25	2	5	36	12	17	21	14	10 (3.3)	31 (8.9)	24 (7.9)
CC1/63	63	Sweden ^{15A-25}	2014	14	0.25	2	5	36	12	17	21	14	11 (3.6)	29 (8.4)	25 (9.3)
CC1/63	4532	SLV of ST 63	2014	14	0.25	2	5	36	1	17	21	14	9 (3)	10 (2.9)	25 (9.3)
CC1/63	63	Sweden ^{15A-25}	2012	18A	0.25	2	5	36	12	17	21	14	10 (3.3)	29 (8.4)	17 (6.6)
CC1/63	63	Sweden ^{15A-25}	2014	14	2.0	2	5	36	12	17	21	14	11 (3.6)	36 (10.4)	24 (9.3)
CC1/63	63	Sweden ^{15A-25}	2014	14	2.0	2	5	36	12	17	21	14	32 (10.6)	43 (12.4)	29 (11.2)
CC5/473	473	SLV of 13079	2013	6A	0.25	7	25	4	4	15	20	28	12 (4)	25 (7.22)	10 (3.8)
CC5/473	13079	SLV of 473	2012	6A	0.5	7	25	4	4	15	739	28	9 (3)	25 (7.22)	9 (3.5)
ST9428	9428	TLV of ST217 Sweden ¹⁻²⁷	2012	19F	0.5	15	5	4	1	17	19	9	1 (0.3)	1 (0.2)	1 (0.3)

^aTPD region.

DLV, double locus variant; FG, founding genotype; Pen, penicillin; PMEN, Pneumococcal Molecular Epidemiology Network; SLV, single locus variant.

and cefotaxime MIC $>2\mu\text{g/mL}$, was also not observed in any of the study isolates. M339F especially when in combination with T338A decreases acylation efficiency to 3- to 10-fold as observed in various studies.^{21,22,28,30} Another major substitution that reduces the binding affinity, S571A/P and Y536F if present together with T338A, seen in Turkey isolates was also not observed in this study isolates. The Q552E mutation was present in both penicillin susceptible and resistant strain as reported earlier by Bicmen *et al.*²⁹ When cefotaxime MIC is between 0.06 and 0.12 $\mu\text{g/mL}$, PBP2x mutations are minimal and present as scattered in individual isolates. As when the cefotaxime MIC increases above 0.25 $\mu\text{g/mL}$, the number and type of aa mutations in PBP2x are accumulated, whereas no unique pattern corresponding to the MIC is observed in the study isolates.

In PBP2b, amino acid substitution T446A (next to one active site S443SN) known to reduce penicillin-binding affinity by 60% was seen in the study isolates with penicillin MIC $>0.25\mu\text{g/mL}$. When penicillin MIC increases above 0.5 $\mu\text{g/mL}$, substitutions such as S412P, N422Y, T426K, and Q427A/L are observed. These variations are present in the “lower lip” of the PBP2b active site and known to decrease the reactivity to beta lactams.²⁸ The reported upper lip substitutions A624E, D625G, and Q628E are present individually in isolates with MIC above 0.12 $\mu\text{g/mL}$, whereas it is seen together and in combination with the lower lip substitutions in isolates with MIC $>1.0\mu\text{g/mL}$ (see Supplementary Table S5). The other “upper lip” substitutions such as A619G is found only in two isolates, and T630N was absent in all the isolates. In PBP2b of PNSP isolates, B3153, H123, and B33586, where the above mutations are not present, it is compensated by mutations in PBP2x and PBP1a. Subsequently, the mutations in PBP2x reduce affinity for cefotaxime resulting in increase of cefotaxime MIC. The PBP2x of isolates with MIC of 0.06 $\mu\text{g/mL}$ had mutations scattered, that is, the individual mutations are distributed among the isolates. When cefotaxime MIC increases, the isolates accumulate the entire variety of entire mutations in PBP2x (see Supplementary Table S6). In isolates B8022 and B1020527, mutations in PBP2x are <3 numbers, which is compensated by mutations in PBP1a.

The major PBP1a mutations observed in isolates with penicillin MIC $\geq 0.25\mu\text{g/mL}$ associated with penicillin resistance are T371A in STMK motif P432T next to SRN motif. A group of four adjacent substitutions in positions 574–577 (TSQF to NTGY), which usually confers intermediate to high-level penicillin resistance (MIC >0.25 – $2.0\mu\text{g/mL}$), were seen in the *pbp1a* gene of the study isolates. This is in concordance with the findings reported by Klugman and colleagues³¹ and Chewapreecha *et al.*³² Observation of combination of all these mutations in study isolates with penicillin MIC $\geq 2.0\mu\text{g/mL}$ and cefotaxime MIC $\geq 1.0\mu\text{g/mL}$ differed from the findings of Zhou *et al.*²²

Thus, in the study isolates, amino acid substitutions are seen in one of the three active sites of one of the three *pbp* genes and also seen around the active sites. The major mutations in and around the active site such as *pbp2b*-T446A, E476G, T489A, V538D AIDTK, 565SVES/T K 569^{24,33} *pbp2x*-T338A, H394L, Q552E and L546V^{28,24} *pbp1a*-T371A, P432T, N₅₇₄TGY, and E388D^{7,34} were found in isolates from meningeal and nonmeningeal disease. No

significant difference in the clinical course or mortality of diseases caused by penicillin nonsusceptible or susceptible pneumococci were observed by Deeks *et al.*, Cardoso *et al.*, and Tan *et al.*^{35–37} Hence, minor changes in the active sites of the three *pbp* genes combined with mosaicism in *pbp1a* gene caused a reduced penicillin susceptibility that increased the MIC values above 0.06 $\mu\text{g/mL}$, which according to meningeal breakpoints denote resistance.

The amino acid variations in PBP2x and PBP1a are high in CC320 isolates followed by CC230 and then CC63 (see Table 2). The PMEN clones associated with these CCs are Taiwan^{19F}-14 (ST236), Denmark¹⁴-32 (ST230), and Sweden^{15A}-25 (ST63), respectively. Figure 2 suggests that majority of the PNSP isolates fall into these CCs, which consists of single locus variants or double locus variants (SLV or DLV) of the PMEN clones associated with each CC. These are the most successful PNSP clones with penicillin MICs between 0.06 and 2.0 $\mu\text{g/mL}$ by 2001.³⁸ All the study isolates had penicillin MIC value between 0.12 and 2.0, which is similar to the MIC of PMEN clones as of 2001. This suggests that PMEN clones remain same or have not undergone major recombinations that increase the penicillin resistance. Sequence similarity $>95\%$ was observed between the PBP1a and PBP2x sequences of SLV present in the study isolates of CC473, between ST9435 and ST2016, and CC320, between ST271 and ST320.

The five isolates with high MIC values had less than five mutations. Among these, the three *pbp* genes of three isolates, B7396, B7765, and B11342, showed no mutations in the conserved motif and had less than three mutations in TPD domain. The penicillin MIC for these isolates were 0.12, 0.25, and 1.0 $\mu\text{g/mL}$; and cefotaxime MIC were 0.25, 0.5, and 0.25 $\mu\text{g/mL}$, respectively. This variation in MIC could be explained by the recent observation that other non-*pbp* genes such as *murM* also influence the expression of penicillin resistance in *S. pneumoniae*.^{39,40} One isolate (B23021) has no mutation in PBP2x, two mutations N538D and G597E, and E388D and S540T-PBP1a had cefotaxime MIC of 0.06 $\mu\text{g/mL}$ and penicillin MIC of 0.12 $\mu\text{g/mL}$.

Among the penicillin nonsusceptible isolates, the majority are serotypes included in pneumococcal vaccines, such as 14, 6B, and 19A/F. These serotypes are known to colonize the nasopharynx of healthy carriers for a long time, which predisposes them to intra-/interspecies recombination that may potentially lead to the development of antibiotic resistance.^{41,42} This can be inferred from the clustering of high MIC study isolates along with sequences of *S. oralis*, *S. infantis*, and *S. mitis* in the phylogenetic tree of *pbp2x* sequences.⁸

Conclusion

This study is first to investigate on the amino acid variations in the *pbp* gene sequences and their relationship with clonal complexes of invasive penicillin nonsusceptible pneumococci isolated from children in India. Our study reveals that the PBP amino acid variations observed in India is similar with those reported worldwide. The alterations observed in this study result only in low-level resistance to beta lactams, which will be resistant according to meningeal criteria. Further studies on a large number of penicillin nonsusceptible isolates are required to improve the understanding of PBP-mediated resistance.

Disclosure Statement

No competing financial interests exist. All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest.

Funding Information

This work was supported by Christian Medical College, Institutional Review Board IRB Min. No: 8200.

Supplementary Material

Supplementary Table S1
Supplementary Table S2
Supplementary Table S3
Supplementary Table S4
Supplementary Table S5
Supplementary Table S6

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